a
Modules: change, group difference and links to growth
Green-purple: limma r-equivalent (contrasts; HR - LR slope); blue-red: Spearman rho within each group of 8. Outline: solid FDR < 0.05, dashed p < 0.05
Training
Acute
HR - LR
Baseline
Predicts gain
Tracks gain
no set at FDR < 0.05 (turquoise)
mRNA3 end processing (blue)
no set at FDR < 0.05 (brown)
no set at FDR < 0.05 (yellow)
Cytoskeleton organization (green)
Aerobic respiration and respiratory e... (red)
Striated muscle contraction (black)
Carbohydrate catabolic process (pink)
Cristae formation (magenta)
Response of EIF2AK4 GCN2 to amino aci... (purple)
300 200 100 0
proteins
HR
LR
HR - LR
T1
T2
T3
fCSA HR - LR
mCSA HR - LR
press HR - LR
fCSA HR
fCSA LR
mCSA HR
mCSA LR
press HR
press LR
FDR < 0.05
p < 0.05
Spearman rho (within group)
1.0 -0.5 0.0 0.5 1.0
r-equivalent (contrast, slope)
0.50 -0.25 0.00 0.25 0.50
b Cristae formation (magenta), person by person
Exploratory: the smallest eigengene p outside the interactions
Acute HR - LR logFC -0.16, p = 0.031, FDR = 0.175
Thin lines: people; thick: group mean
HR
LR
eigengene
0.2
0.0
-0.2
-0.4
T1
T2
T3
c Module preservation between groups
Each group's WGCNA modules tested in the other group's data;
Zsummary < 2 none, > 10 strong. Row: group module → full-cohort module holding most of its proteins
yellow → red (69%)
pink → black (69%)
black → pink (63%)
blue → turquoise (54%)
turquoise → blue (48%)
greenyellow → brown (63%)
purple → brown (54%)
salmon → purple (60%)
red → brown (48%)
brown → turquoise (59%)
magenta → blue (62%)
green → blue (37%)
tan → green (40%)
black → blue (58%)
green → black (60%)
midnightblue → blue (77%)
magenta → green (75%)
brown → turquoise (39%)
red → purple (32%)
turquoise → brown (58%)
pink → blue (22%)
greenyellow → red (31%)
yellow → yellow (44%)
blue → yellow (28%)
salmon → red (30%)
lightcyan → blue (46%)
tan → pink (56%)
purple → brown (26%)
cyan → red (30%)
HR modules in LR
LR modules in HR
0
2
5
10
Zsummary
proteins
100
200
300